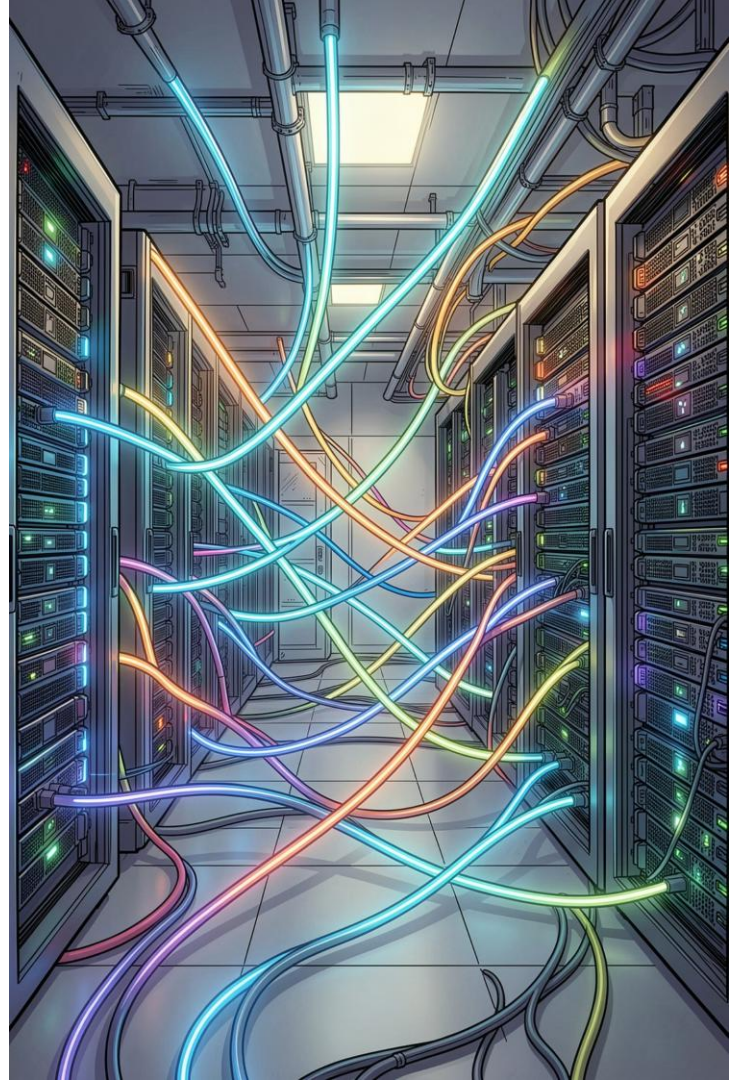


AEGIS RAG

An Enterprise-Grade Secure, Multi-Source Retrieval-Augmented Generation System with Strict Role-Based Access Control and NLI-Based Hallucination Mitigation

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The Problem

Enterprise Data is Siloed — and Dangerous to Expose



Modern enterprises store critical data across disconnected silos — PDFs, SQL, CSV, JSON — each with its own access policies. Employees need natural language querying, but must never see data outside their clearance.

LLMs Alone

Hallucinate and ignore access restrictions

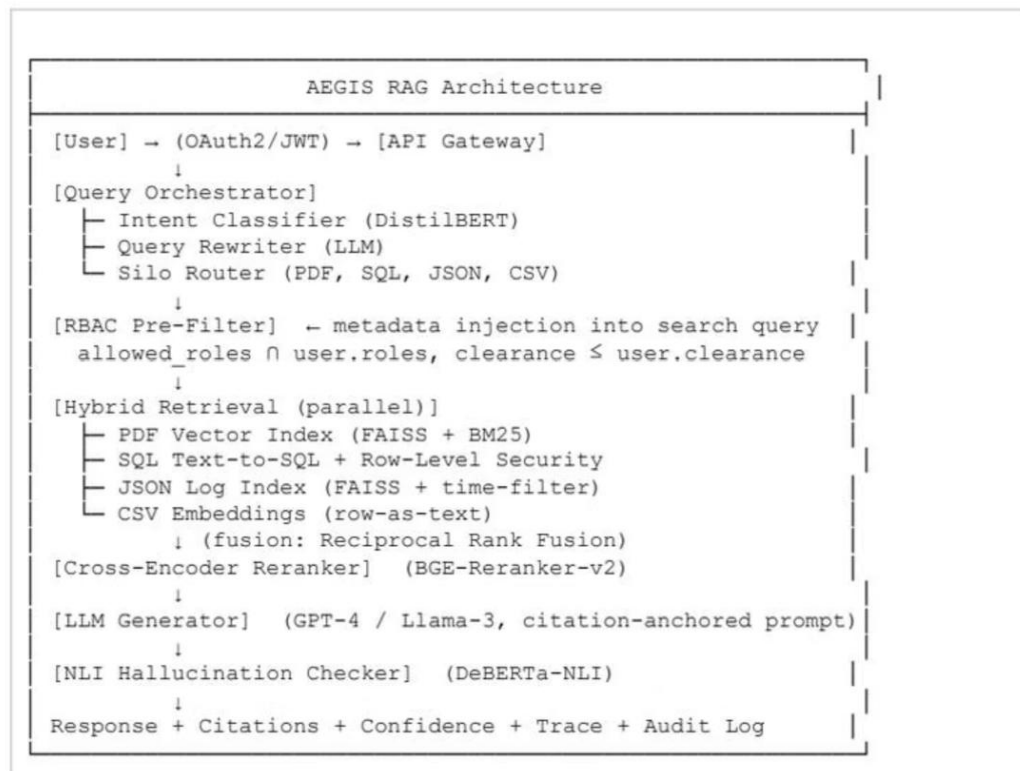
Traditional Search

Lacks semantic understanding and fine-grained security

Existing RAG

No robust RBAC, multi-source integration, or hallucination prevention

The 7-Stage AEGIS RAG Pipeline



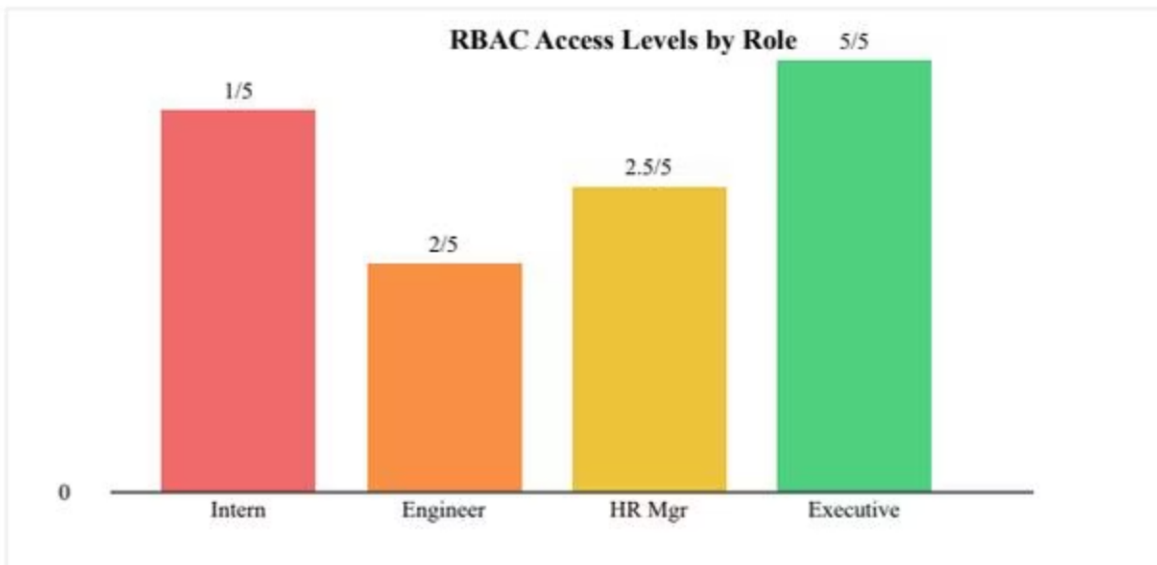
Every query flows through authentication, intent classification, RBAC pre-filtering, multi-source hybrid retrieval, cross-encoder reranking, grounded LLM generation, and NLI hallucination verification — all stages logged immutably.

Zero Data Leakage — By Design

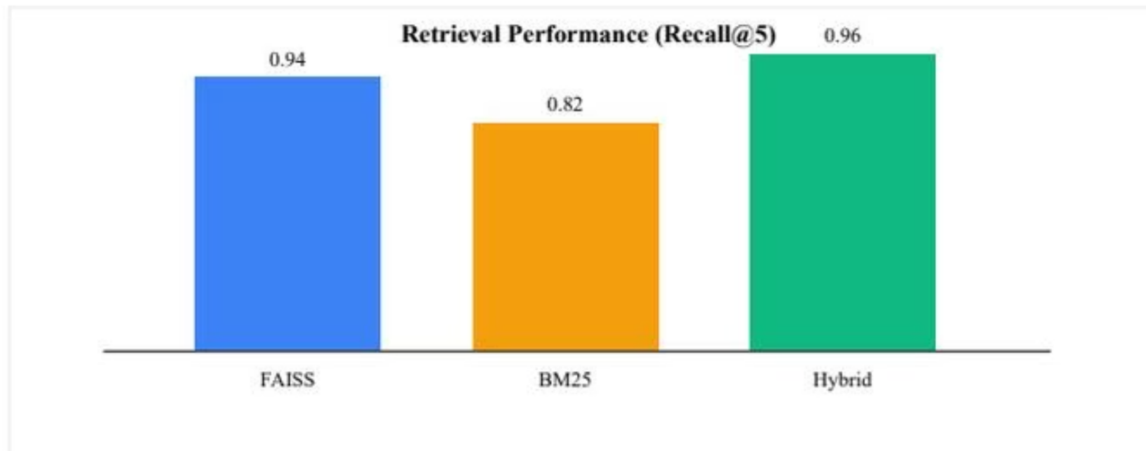
Each user's JWT encodes roles, department, and clearance level. Before any similarity search, AEGIS injects a boolean pre-filter ensuring the LLM **never** receives **unauthorized text**:

i $\text{visible}(\text{chunk}) \iff$
 $\text{chunk.allowed_roles} \cap \text{user.roles} \neq \emptyset \wedge \text{chunk.min_clearance} \leq \text{user.clearance}$

Implemented via PostgreSQL JSONB containment operators with GIN indexing — $O(1)$ complexity. Zero RBAC bypass incidents recorded across all experiments.



Hybrid Dense + Sparse Retrieval



Scores from both retrievers are fused via weighted sum ($\alpha = 0.6$), then top-20 candidates are reranked by a BGE-Reranker-v2 cross-encoder. Final top-5 chunks are passed to the generator.

FAISS IVF

Dense vector search · Recall@5:
0.94

BM25 (TF-IDF)

Sparse keyword scoring · Recall@5:
0.82

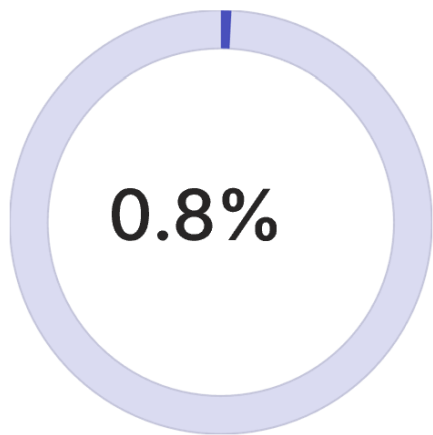
Hybrid

Fused + reranked · Recall@5: **0.96**

NLI Verification Cuts Hallucinations to Under 1%

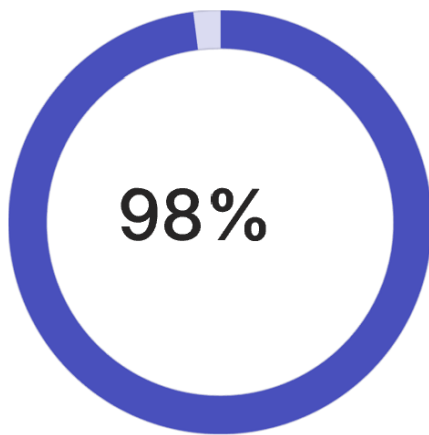
After generation, every factual claim is paired with the top-5 retrieved chunks and scored by a fine-tuned DeBERTa-NLI model. A claim is "supported" if entailment score exceeds 0.7. If NLI Score < 0.75, the response is flagged for human review.

$$NLI_Score = \frac{| \text{supported claims} |}{| \text{total claims} |}$$



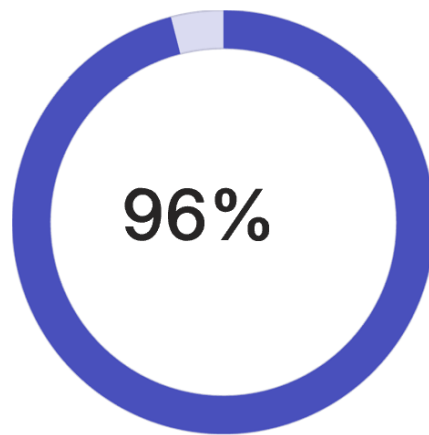
Hallucination Rate

Down from 5.2% without NLI



Factual Consistency

Human eval, up from 91%

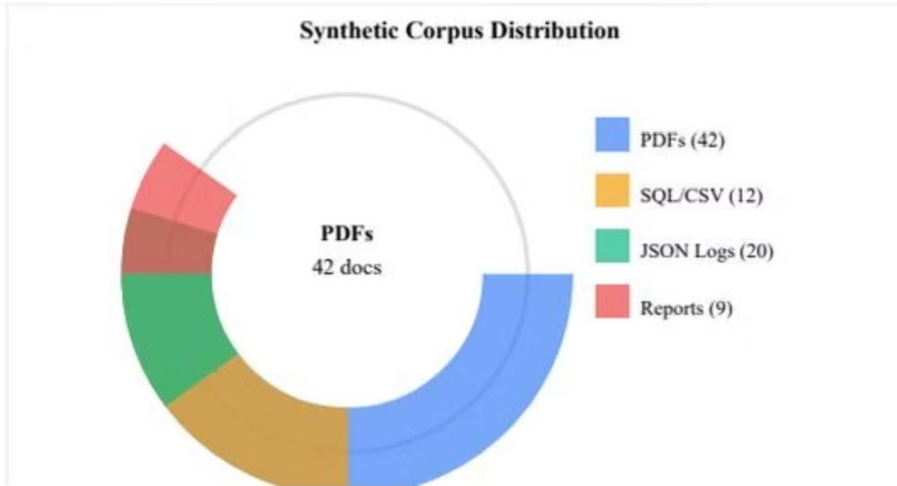


Avg NLI Score

Across all authorized queries

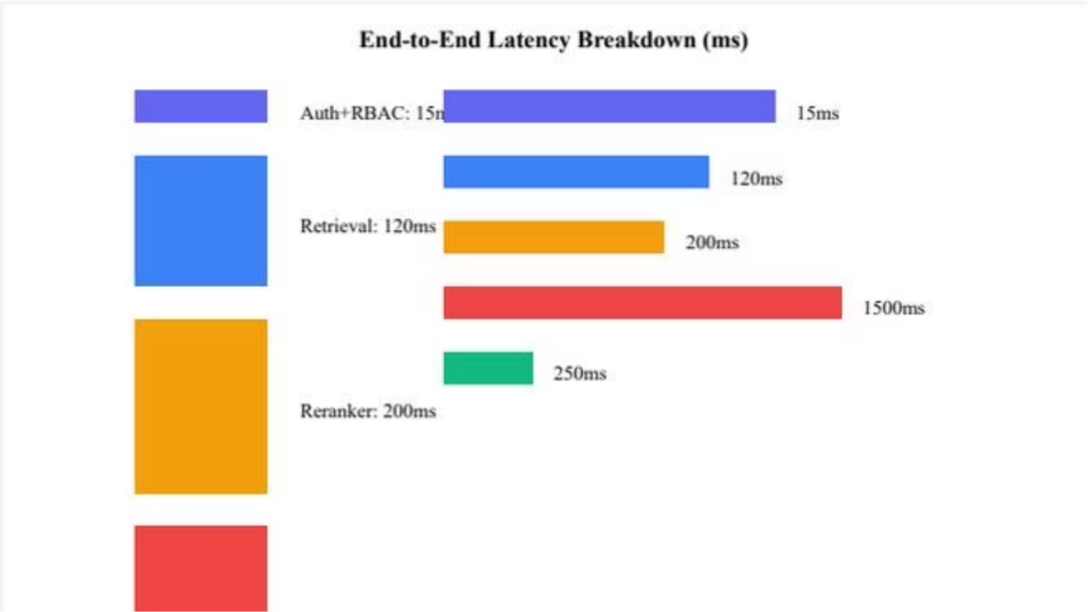
Synthetic Corpus & Technology Stack

83 documents · 12,847 chunks, each annotated with RBAC metadata. Generated using Faker, PyMuPDF, pandas, and SQLAlchemy.



Layer	Technology
Backend	Django 5.0 + DRF
Vector Index	FAISS 1.7.4 IVFFlat
Embeddings	all-MiniLM-L6-v2
LLM	GPT-4 (Azure) / Llama-3-70B
NLI	DeBERTa-v3-small
Deployment	Docker · Kubernetes · Helm

Performance Across the Full Pipeline



Sub-second Retrieval

Auth+RBAC: 15ms · Retrieval: 120ms ·
Reranker: 200ms

96% Confidence

On authorized queries · Recall@5: 0.94 ·
MRR: 0.91

100% RBAC Enforcement

Zero bypass incidents across 200 test
queries

Real-World Query Responses

✓ HR Manager Query

"What are the salary bands for L4 engineers?"
→ \$80k–\$110k annually [c1][c7] · **Confidence: 96%**

✗ Intern Query (Same Question)

"What are the salary bands for L4 engineers?"
→ "Cannot answer — insufficient permissions." · **Confidence: 0%**

✓ Executive Cross-Source Query

"Server uptime and compliance issues?"
→ 99.97% uptime [c4] · SOX AC-4 quarterly reviews [c3] · **Confidence: 94%**



Conclusion

AEGIS RAG is a **production-ready, turnkey enterprise RAG system** — uniquely combining strict RBAC enforcement, hybrid retrieval, grounded generation, and NLI hallucination mitigation in a single deployable solution.

Secure

Zero RBAC bypass · Pre-filter at retrieval level

Accurate

0.8% hallucination rate · 98% factual consistency

Explainable

Citations, confidence scores, immutable audit log

Deployable

Docker · Kubernetes · Full source code included